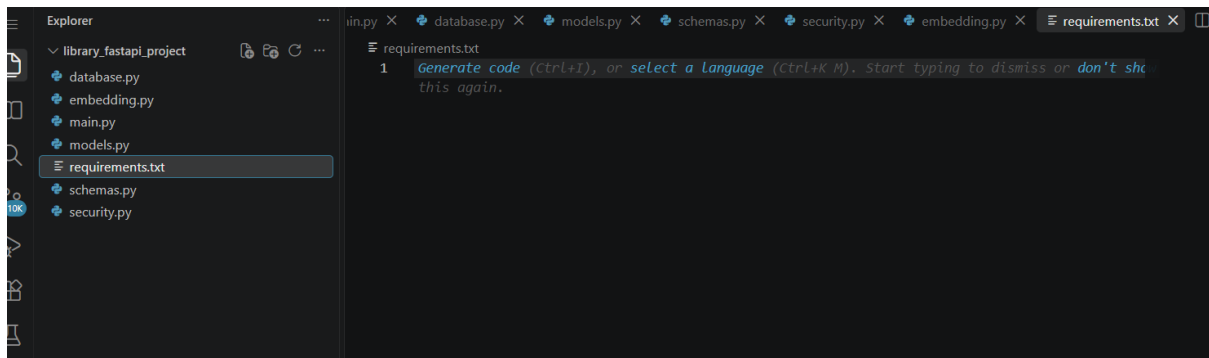
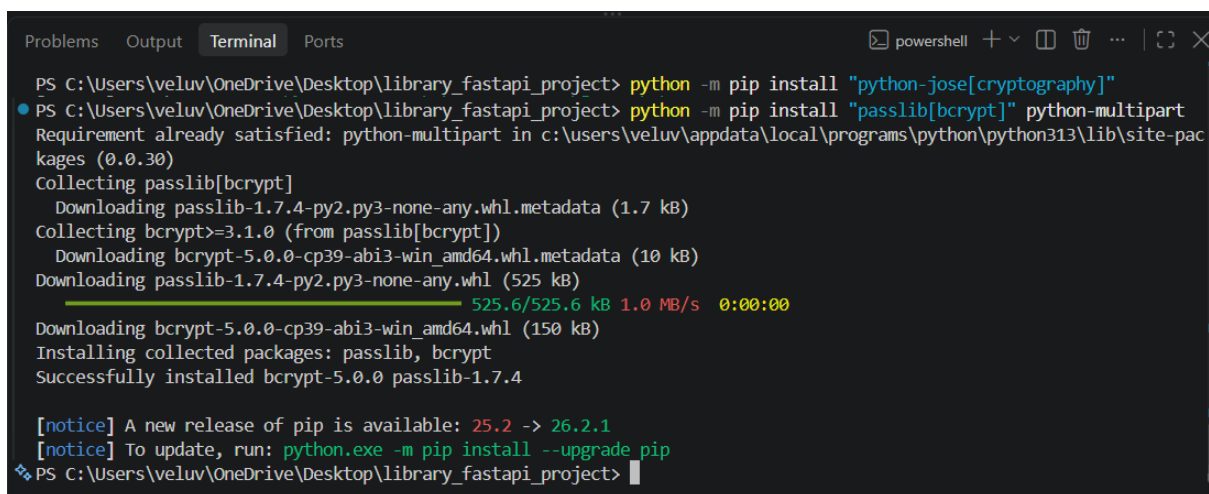


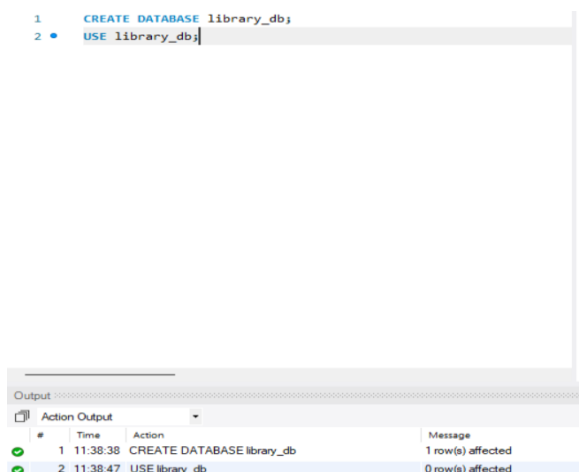
WEEK 5 — LIBRARY FASTAPI PROJECT



STEP 2 — Install the Python packages

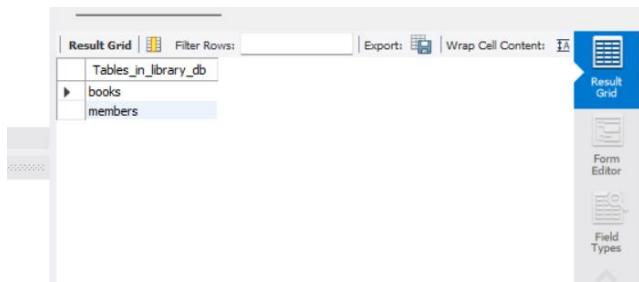


STEP 3 — Create the MySQL Database



Creation and verification of members and books tables in MySQL.

✓	3	11:39:41	CREATE TABLE IF NOT EXISTS members (id IN...	0 row(s) affected	0.047 sec
✓	4	11:40:02	CREATE TABLE IF NOT EXISTS books (id INT ...	0 row(s) affected	0.047 sec



database.py

```

library_fastapi_project
├── .venv
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── database.py > ...
1  from sqlalchemy.ext.asyncio import (
2      AsyncSession,
3      async_sessionmaker,
4      create_async_engine
5  )
6  from sqlalchemy.orm import DeclarativeBase
7
8  DATABASE_URL = (
9      "mysql+asyncmy://root:Bhardwaj@2025"
10     "@localhost:3306/library_db"
11 )
12
13 engine = create_async_engine(DATABASE_URL, echo=True)
14
15 AsyncSessionLocal = async_sessionmaker(
16     engine,
17     expire_on_commit=False

```

Screenshots

```

embedding.py > text_to_vector
1  import hashlib
2  import math
3
4  VECTOR_DIM = 384
5
6
7  def text_to_vector(text: str) -> list[float]:
8      values = []
9
10     for i in range(VECTOR_DIM):
11         digest = hashlib.sha256(
12             f"{text.lower():{i}}".encode()
13         ).digest()
14
15         number = int.from_bytes(
16             digest[:4],
17             byteorder="big"
18         )
19

```

```

security.py > get_current_member
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE THIS SECRET"

```

```

schemas.py > TokenResponse
1  from datetime import date
2  from decimal import Decimal
3
4  from pydantic import BaseModel, Field, field_validator
5
6
7  class BookSchema(BaseModel):
8      title: str

```

Swagger UI

default

GET

/ Home

POST

/register Register

POST

/login Login

POST

/books Create Book

GET

/books/search Search Books

Register a Library Member

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Bhardwaj",
    "password": "library123"
  }'
```

Request URL

```
http://127.0.0.1:8000/register
```

Server response

Code	Details
500	Error: Internal Server Error

L

Login & Generate JWT Token

200	Successful Response	No links
Media type		
application/json		
Controls Accept header.		
Example Value Schema		
<pre>{ "access_token": "string", "token_type": "string" }</pre>		
422	Validation Error	No links
Media type		
application/json		

Authorize Swagger

Available authorizations

X

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.

API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: /login

Flow: password

username: testuser

password: *****

Client credentials location: basic

client_secret: *****

Logout

Close